a ladder activity

Equations of Parallel and Perpendicular Lines

MRS E. TEACHES MATH

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A special thank you to the following artists:

Instructions:

- The two following "worksheet" pages should be copied single-sided and given to each student or group of students. I like to have students work in partners for this activity.
- Students will work the problems in the space provided.
- When they are finished, students will cut along the lines to make a total of 12 squares.
- The first card says "Start". Students will then match that card to the answer that is on the top of another card. They will continue in this fashion, making a chain. They will tape their completed chain together.

Suggestions:

- I like to copy the problems on colored paper and then hang a few of the completed chains in my classroom as decoration.
- These cards could be copied on cardstock and laminated for reuse.
- If you're concerned about the amount of paper, this could be copied two per page. The squares will just be smaller.
- Solutions are included at the end of the document.

Start

Write an equation of the line with y-intercept 3 that is parallel to $y=-\frac{3}{4}x-4$.

$y=-3x+5$

Write an equation of the line with y-intercept -2 that is perpendicular to $y=4x+6$.

$y=-\frac{3}{4}x+3$

Write the equation of a line that is perpendicular to the line $3x+2y=8$ and passes through the y-intercept of that line.

$y=\frac{1}{2}x-2$

Write the equation for the line that passes through $(4,7)$ and is perpendicular to the line $y=\frac{2}{3}x-1$.

$y=-\frac{5}{3}x+1$

Write an equation of the line that is parallel to $y=-3x+2$ and passes through the point $(2,1)$.

$y=-\frac{1}{4}x-2$

Line j passes through the points $(2,1)$ and $(0,0)$. Line k passes through the point $(4,3)$ and is parallel to line j. Find an equation for line k.

$y = 7x$

End

$y = \frac{2}{3}x + 4$

Line j passes through the points $(-5,-2)$ and $(5,4)$. Line k passes through the point $(-3,6)$ and is perpendicular to line j. Find an equation for line k.

$y = -\frac{3}{2}x + 13$

Write the equation of a line that is perpendicular to the line $5x - 3y = 12$ and passes through the point $(3,1)$.

$y = \frac{1}{2}x + 1$

Write the equation for the line that is parallel to the line $y = 7x - 3$ and passes through the origin.

$y = -3x + 7$

Write an equation of the line that passes through the point $(4,0)$ and is perpendicular to $y = -2x + 1$.

$y = -\frac{3}{5}x + \frac{14}{5}$

Line j passes through the points $(-4,10)$ and $(2,-8)$. Line k passes through the point $(1,2)$ and is parallel to line j. Find an equation for line k.

Writing Equations of Lines Chain
Solutions

1. Start
2. $y = -\frac{3}{4}x + 3$
3. $y = \frac{2}{3}x + 4$
4. $y = -\frac{5}{3}x + 1$
5. $y = -3x + 7$
6. $y = \frac{1}{2}x - 2$
7. $y = -\frac{3}{2}x + 13$
8. $y = -\frac{3}{5}x + \frac{14}{5}$
9. $y = -3x + 5$
10. $y = -\frac{1}{4}x - 2$
11. $y = \frac{1}{2}x + 1$
12. $y = 7x$